

Lesson 9 Root space decomposition

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Thm s.s. Linear Lie algebra: usual and abstract Jordan decomposition coincide

Let $L \subset \mathfrak{gl}(V)$ be semisimple (V f.d.). Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and the usual Jordan decompositions in L coincide.

$a = a_s + a_n \Rightarrow \text{ad}_{\mathfrak{gl}(V)}(a) = \text{ad}_{\mathfrak{gl}(V)}(a_s) + \text{ad}_{\mathfrak{gl}(V)}(a_n)$
Then $\text{ad}_L a_s$ s.s. & $\text{ad}_L(a_n)$ nilpotent
 $\Rightarrow \text{ad}_L(a) = \text{ad}_L(a_s) + \text{ad}_L(a_n)$

pf:

For any $a \in L$, let $a = a_s + a_n$ be its Jordan decomposition in $\mathfrak{gl}(V)$. Then $\text{ad}(a) = \text{ad}(a_s) + \text{ad}(a_n)$ is the Jordan decomposition of $\text{ad}_{\mathfrak{gl}(V)}(a)$ in $\mathfrak{gl}(\mathfrak{gl}(V))$.

We have $\text{ad}(a_s) = \psi(\text{ad}(a))$ for some polynomial ψ with no constant term. Then $[a, L] \subset L \Rightarrow [a_s, L] \subset L$ i.e. $x \mapsto [a_s, x]$ is a derivation of L . Since L is s.s., any derivation is inner, thus $\exists a_s' \in L$ s.t. $\forall x \in L, [a_s, x] = [a_s', x]$, i.e. $[a_s - a_s', L] = (0)$.

Now, write $a - a_s' = (a_s - a_s') + a_n$; write $V = \bigoplus V_i$ V_i irreducible
 $\uparrow$ $\uparrow$
 L nilpotent.

On one hand, $[a_s - a_s', L] = (0) \Rightarrow a_s - a_s'$ acts by scalars on V_i .

On the other hand, since $L = [L, L]$, $\text{tr}_{V_i}(a - a_s') = 0$; $\text{tr}(a_n) = 0$ as a_n is nilpotent $\Rightarrow \text{tr}_{V_i}(a_s - a_s') = 0 \Rightarrow a_s - a_s'$ acts trivially on V_i
 $\Rightarrow a_s = a_s' \Rightarrow a_s \in L \Rightarrow a_n \in L$.

Corollary Jordan decomposition in representations

Let L be a semisimple Lie algebra, $\phi: L \rightarrow \mathfrak{gl}(V)$ a finite dimensional representation of L . If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, then $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.

pf: By above theorem, it suffices to show $\text{ad}_{\phi(L)} \phi(x) = \text{ad}_{\phi(L)} \phi(s) + \text{ad}_{\phi(L)} \phi(n)$ is the abstract Jordan decomposition of $\text{ad}_{\phi(L)} \phi(x)$.

Indeed, $\phi(L)$ is spanned by the eigenvectors of $\text{ad}_{\phi(L)} \phi(s)$ as L is w.r.t. ad_s , thus $\text{ad}_{\phi(L)} \phi(s)$ is semisimple. Similarly, $\text{ad}_{\phi(L)} \phi(n)$ is nilpotent and it commutes with $\text{ad}_{\phi(L)} \phi(s)$. $\forall [x, y] \mapsto [\phi(x), \phi(y)] = \phi([x, y])$

Apply the above theorem, we get the desired result.

Root space decomposition

Def toral

A Lie algebra L is called toral if $\forall 0 \neq a \in L$, $\text{ad}(a): L \rightarrow L$ is semisimple.

Lemma Any toral Lie algebra is abelian.

pf:

Let $0 \neq a \in L$. Write $L = \bigoplus_{\alpha} L_{\alpha}$ eigenspaces of $\text{ad}(a)$.

Suppose otherwise $\exists \alpha \neq 0$ s.t. $L_{\alpha} \neq \{0\}$. For any $0 \neq b \in L_{\alpha}$, note that $\text{ad}(b)L_{\beta} \subseteq L_{\beta+\alpha}$, $\text{ad}(b)^2 L_{\beta} \subseteq L_{\beta+2\alpha}$, $\dots$; $\exists k \geq 1$, $\beta+k\alpha$ is not an eigenvalue, i.e. $\exists k$ $\text{ad}(b)^k L_{\beta} = \{0\} \Rightarrow \text{ad}(b)$ is both semisimple and nilpotent $\Rightarrow \text{ad}(b) = 0$.

Let L be semisimple and let H be a maximal Lie subalgebra of L s.t.

$\forall 0 \neq h \in H$, $\text{ad}_L(h)$ is semisimple. existence of H : if $\text{ad} x$ is nilpotent $\forall x \in L$,

Write $L = L_0 + \sum_{\alpha \in H^*} L_{\alpha}$ eigenspaces L is nilpotent by Engel's thm.

notation: $\forall h \in H$, $\alpha \in H^$, $\alpha \in L_{\alpha}$,

$$\begin{aligned} \text{ad}(h)\alpha &= \alpha(h)\alpha \\ &\quad \text{"} \\ &\quad \langle \alpha, h \rangle \\ &\quad \text{"} \\ &\quad \langle h, \alpha \rangle \end{aligned}$$

Denote $\Delta = \{ \text{nonzero eigenfunctions} \} \Rightarrow L = L_0 + \sum_{\alpha \in \Delta} L_{\alpha}$

Lemma For any $\alpha, \beta \in \Delta \cup \{0\}$, if $\alpha + \beta \neq 0$, then $(L_{\alpha} | L_{\beta}) = \{0\}$.

pf: $\exists 0 \neq h \in H$, s.t. $\langle \alpha + \beta, h \rangle \neq 0$.

For any $0 \neq \alpha \in L_{\alpha}$, $0 \neq \beta \in L_{\beta}$, we have $([\alpha, h] | \beta) = (\alpha | [h, \beta])$

where $([\alpha, h] | \beta) = -\langle \alpha, h \rangle (\alpha | \beta) \Rightarrow (\alpha | \beta) = 0$.

$$(\alpha | [h, \beta]) = \langle \beta, h \rangle (\alpha | \beta)$$

Lemma $L_0 = H$

pf: $L_0 = C_L(H)$

For any $0 \neq a \in L_0$, let $a = a_s + a_n$ be its abstract Jordan decomposition $\text{ad}(a) = \text{ad}(a_s) + \text{ad}(a_n)$. Then $a_s, a_n \in L_0 \Rightarrow H + F a_s$ acts s.s. Bu

For any $0 \neq a \in L_0$, let $a = a_s + a_n$ be its abstract Jordan decomposition $\text{ad}(a) = \text{ad}(a_s) + \text{ad}(a_n)$. Then $a_s, a_n \in L_0 \Rightarrow H + F a_s$ acts s.s. By maximality of H , $a_s \in H$.

Claim: $a_n = 0 \Rightarrow a = a_s \in H$

We need to show that $(a_n | L_0) = \{0\}$ (From above lemma, $(\cdot | \cdot)$ restricted to L_0 is nondegenerate, i.e. if $x \in L_0$ $(x | L_0) = 0 \Rightarrow x = 0$).

Note that $\forall a \in L_0$, $a_s \in H$, thus $\text{ad } a | L_0 = \text{ad } a_s | L_0 + \text{ad } a_n | L_0 = \text{ad } a_n | L_0$.

i.e. $\forall a \in L_0$, $\text{ad}_L a$ is nilpotent $\Rightarrow L_0$ is nilpotent by Engel's thm.

In particular, L_0 is solvable, thus by Lie's thm, $\exists$ a basis of L s.t. the matrices of L_0 relative to this basis are upper triangle. Since $\text{ad } a_n$ is nilpotent, $\forall x \in L_0$, $(a_n | L_0) = 0$.

$$(\text{tr}(\text{ad } a_n \text{ ad } x) = 0)$$

$$\begin{matrix} \downarrow & \downarrow \\ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} a_n & * \\ 0 & a_n \end{pmatrix} \end{matrix}$$

Corollary $(\cdot | \cdot)_H$ is nondegenerate. $\rightarrow$ Similarly, $\forall \alpha \in \Delta$, $\forall \alpha \in L_\alpha$, $(\alpha | L_{-\alpha}) = 0 \Rightarrow \alpha = 0$.
(i.e. $\forall h \in H$, $(h | H) = 0 \Rightarrow h = 0$).

We may then identify H with H^* through

$$v: H \rightarrow H^* \quad \text{with} \quad v^{-1}: H^* \rightarrow H$$

$$h \mapsto (h | \cdot)_H$$

$$\forall \alpha \in H^*, \quad \alpha(y) = (v^{-1}(\alpha) | y). \\ \langle \alpha, y \rangle$$

$$\text{i.e. } v(h)(y) = (h | y) \quad \forall y \in H$$

$$\langle v(h), y \rangle$$

Moreover, for any $\alpha, \beta \in H^*$, we can define $\alpha(\beta) = (v^{-1}(\alpha) | v^{-1}(\beta))$

Lemma

$$\forall \alpha \in L_\alpha, \gamma \in L_{-\alpha}, \alpha \in \Delta$$

$$[\alpha, \gamma] = (\alpha | \gamma) v^{-1}(\alpha)$$

$$\text{pf: } ([h, \alpha] | \gamma) = (h | [\alpha, \gamma])$$

$$\langle \alpha, h \rangle (\alpha | \gamma) = (v^{-1}(\alpha) | h) (\alpha | \gamma)$$

$$\langle \alpha, h \rangle (x_\alpha | y_{-\alpha}) = (v^{-1}(\alpha) | h) (x_\alpha | y_{-\alpha}) \\ = (h | (x_\alpha | y_{-\alpha}) v^{-1}(\alpha))$$

Since h is arbitrary, $[x_\alpha, y_{-\alpha}] \in L_0 = \mathfrak{h}$, we conclude that $[x_\alpha, y_{-\alpha}] = (x_\alpha | y_{-\alpha}) v^{-1}(\alpha)$

Lemma

$$\forall \alpha \in \Delta, (\alpha | \alpha) \neq 0.$$

pf:

From above lemma, $\forall x_\alpha \in L_\alpha$, choose $y_{-\alpha} \in L_{-\alpha}$ s.t. $[x_\alpha, y_{-\alpha}] = (x_\alpha | y_{-\alpha}) v^{-1}(\alpha) \neq 0$.
Suppose otherwise that $(\alpha | \alpha) = 0 \Rightarrow \langle \alpha, v^{-1}(\alpha) \rangle = (v^{-1}(\alpha) | v^{-1}(\alpha)) = 0$

Consider $L' = \mathbb{F}x_\alpha + \mathbb{F}y_{-\alpha} + \mathbb{F}v^{-1}(\alpha) \Rightarrow L'$ is nilpotent ($[v^{-1}(\alpha), x_\alpha] = \alpha(v^{-1}(\alpha))x_\alpha = 0$)
Since $L' \cong \text{ad}_L L'$, in particular, $\text{ad}_L L'$ is solvable, by Lie's thm, $v^{-1}(\alpha) \in [L' L']$ acts nilpotently on L . Thus $v^{-1}(\alpha)$ is both semisimple and nilpotent $\Rightarrow v^{-1}(\alpha) = 0$, a contradiction.

For any $\alpha \in \Delta$, $\exists x_\alpha \in L_\alpha, y_{-\alpha} \in L_{-\alpha}$, s.t. $(x_\alpha | y_{-\alpha}) = \frac{2}{(\alpha | \alpha)}$; $h_\alpha = \frac{2}{(\alpha | \alpha)} v^{-1}(\alpha) \in \mathfrak{h}$.
Then $(x_\alpha, y_{-\alpha}, h_\alpha)$ is an $\mathfrak{sl}_2$ -triple:

$$[h_\alpha, x_\alpha] = \alpha(h_\alpha) x_\alpha = \frac{2}{(\alpha | \alpha)} (v^{-1}(\alpha) | v^{-1}(\alpha)) x_\alpha = 2x_\alpha \\ [h_\alpha, y_{-\alpha}] = -\alpha(h_\alpha) y_{-\alpha} = \frac{2}{(\alpha | \alpha)} (v^{-1}(\alpha) | v^{-1}(\alpha)) y_{-\alpha} = -2y_{-\alpha} \\ [x_\alpha, y_{-\alpha}] = (x_\alpha | y_{-\alpha}) v^{-1}(\alpha) = h_\alpha$$